

Case Study 11: Human Activity Recognition

VI^o Problem

→ Smartphone → Sensors:-

- 1) Accelerometer → Acceleration
- 2) Gyroscope → Angular Velocity.

Determine:-

- 1) Walking
- 2) Upstairs walking
- 3) Downstairs "
- 4) Sitting
- 5) Standing up
- 6) Laying

Why is this useful? (In watch)

- | | |
|--------------------------------|-------------------|
| 1) Burnt Calories Calculation | } Health Measures |
| 2) Calories burnt " | |
| 3) Heart-Rate " | |
| 3) Sleep Duration | |

Accelerometers are Tri-axial (3 axis)
Gyroscope also " " (")

Input:

You have time series data for both sensors for all 3 axis.

So total 6 time series data.

Output: Predict

So it is a multiclass classification problem.

Measures:

- 1) log loss
- 2) Confusion matrix
- 3) Multi class log loss.

• V2: Dataset Understanding

Time is divided into window. Each window overlaps each ~~other~~ previous window by 50%. This is called Overlapped Sampling.

We'll apply filter and sampling and convert it into numerical vectors.

So the vector size is 128.

Domain experts have designed features for us out of the ~~team~~ ~~there~~ 6 time series data.

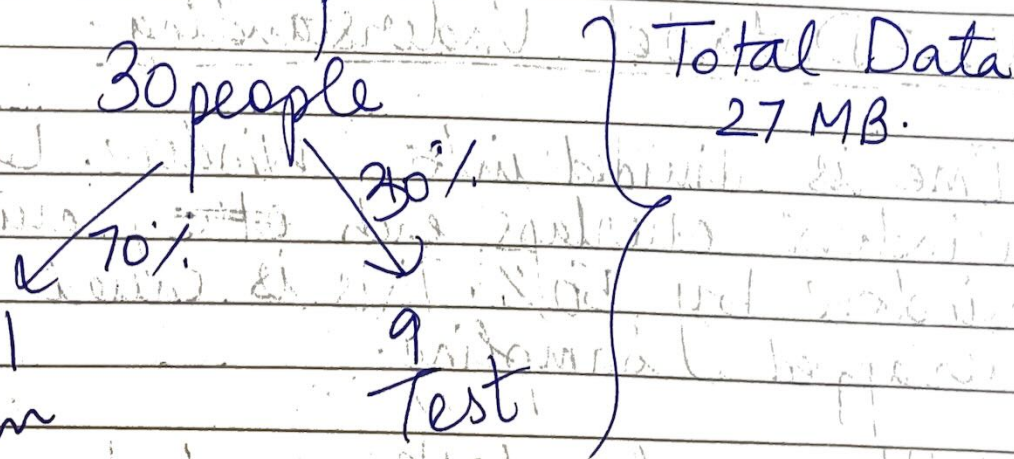
t - time signal.

f - fourier transform.

This case study is a competition
6/10

- 1) Expert engineered features on which Machine learning models will be applied.
- 2) Raw time series on which Deep learning models will be applied.

Train Test Split



Expert generated 561 features.

V3: Data clearing & preprocessing

- For each person, find every class category on plot.
- Datapoints per class.
- changing feature names.

V4: EDA: Univariate Analysis

V5: EDA: Data visualization using t-SNE

We have 561 vector representation of expert generated features.

We apply t-SNE.

1) Perplexity = 2

Max_iter = 1000

Standing and Sitting overlap.

2) Perplexity = 5

max_iter = 1000.

3) Perplexity = 10

max_iter = 1000.

V6: Classical ML models

1) HAR_EDA.ipynb. → EDA file.

2) HAR_PREDICTION_MODELS.ipynb.

- ↳ 561 expert features.
- ↳ Apply classical ML models.
- ↳ Removed special symbols.
- ↳ Loaded the csv files
- ↳ 564 columns.
- ↳ Train/Test split already done.
- ↳ Y label is also separated.
- ↳ plot confusion matrix func.
- ↳ performance model func.
- ↳ grid search func.
 - ↳ Logistic Regression
 - 1, 12 penalty.
 - C - hyperparameter.
 - Acc - 96.26%
 - Confusion Matrix getting plotted.
 - Laying feature predicted 100% acc.
 - Walking also 100% acc.
 - Sitting predicted bad
 - Precision - low for standing
 - Recall - " " Sitting.
 - Best parameter = 30.
 - " Acc - 94.61

↳ Linear SVM with Grid Search.

- Confusion b/w sitting and standing.

↳ Kernel SVM with Grid Search

- Better in terms of pred. of sitting & standing
- RBF Kernel used
- 94.40% Acc

↳ Decision Trees with Grid Search

↳ RF with Grid Search.

↳ Comparison b/w models

↳ Accuracy vs. Error Table.

↳ Better performance with Log. Reg. & Linear SVM.

↳ Deep learning

↳ Automatically engineered features for us.

↳ LSTM in next sec.

V7: Deep Learning Model

model - RMSprop
Metric - Accuracy
categorical_crossentropy = loss

Case Study II: Human Activity Detection

Assignment Tasks

we have seen :-

Simple LSTM + dropout $\rightarrow$ 91%

Classical ML + expert features $\rightarrow$ 96%

1) Hyperparameter Tuning Experiments

i) 32 LSTM units change

Multiple ii) Dropout Rate Instead of 0.5 change

Creates iii) 2-LSTM layers + larger dropouts

1352 datapoints

So that you don't overfit

Get better results than ML models

Conclusion

$\rightarrow$ Step by Step procedure on how you solved this case study.

$\rightarrow$ Compare all models using pretty table